Exercise 7: Financial Forecasting

1. Understanding Recursive Algorithms

Recursion

Recursion is a programming technique in which a method calls itself to solve a problem. It consists of a base case that stops the recursion and a recursive case that reduces the problem into smaller subproblems. Recursion can simplify problems involving repeated calculations and mathematical relationships.

2. Analysis

Time Complexity

The recursive algorithm performs one recursive call for each year.

Time Complexity: O(n)

where **n** represents the number of years.

Optimization

The recursive solution can be optimized by:

- Using memoization to store previously calculated results.
- Using an iterative approach instead of recursion.
- Applying the direct compound growth formula when suitable.

These approaches reduce unnecessary computations and improve efficiency for larger forecasting periods.

Conclusion

Recursion provides a simple and effective method for implementing financial forecasting. However, for larger datasets and longer forecasting periods, optimized techniques such as memoization or iterative solutions are recommended to improve performance and reduce computational overhead.